Data Science Lab

Name: Vishal Rajesh Mahajan Experiment No: 7

Class: BE INFT B-1 Roll No: 1

To implement Supervised Learning algorithm - Ada Boost.

Import Required Libraries

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import seaborn as sns
import matplotlib.pyplot as plt
```

Load Training Data

```
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)
print("Dataset shape:", df.shape)
print(df.head())
```

```
Dataset shape: (891, 12)
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

Preprocessing

```
# Drop unnecessary columns
df = df.drop(['Name', 'Ticket', 'Cabin'], axis=1)

# Fill missing values
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Encode categorical features
le = LabelEncoder()
for col in ['Sex', 'Embarked']:
    df[col] = le.fit_transform(df[col])
```

Split into train/test

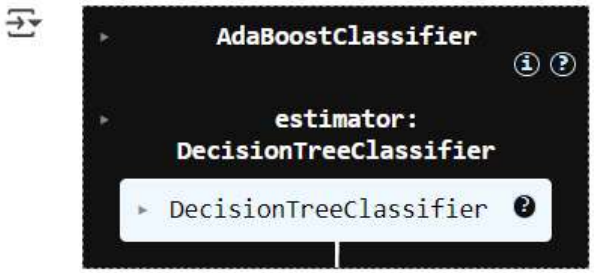
```
X = df.drop('Survived', axis=1)
y = df['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Define AdaBoost model

```
base_model = DecisionTreeClassifier(max_depth=2, random_state=42)
ada_model = AdaBoostClassifier(
    estimator=base_model,
    n_estimators=100,
    learning_rate=0.5,
    random_state=42
)
```

Train model

```
ada_model.fit(X_train, y_train)
```



Make predictions

```
y_pred = ada_model.predict(X_test)
```

Evaluate

```
print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.7877094972067039

Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.85      0.83       110
           1       0.75      0.68      0.71        69

    accuracy                           0.79       179
   macro avg       0.78      0.77      0.77       179
weighted avg       0.79      0.79      0.79       179
```